LOT (R) COMPUTER

SOFTWARE / API CONNECTOR MANUAL -- 3 July 2026

Document: LOT-COMPUTER-SOFTWARE-BRIDGE.md

Related, kept separate on purpose:

docs/technical/LOT-COMPUTER-FIRMWARE.md (device-side firmware)

docs/technical/LOT-COMPUTER-RIG-SPEC.md (hardware)

This document is the **software that connects the firmware to lot-systems.com** — the LOT® API connector. It is kept as its own document because it evolves on the platform's release cadence (frequent, git-deployed), not the firmware's (slow, OTA-signed, riskier). Conflating the two would force platform API changes to wait for a firmware release cycle, or vice versa.

00 — Two Sync Paths

```
LOT® Computer (device)
    ■
    ■■■ Path A: BLE → LOT Bridge companion app (phone/desktop) → HTTPS → lot-systems.com
    ■          Preferred. Lower device power draw (§05, firmware doc).
    ■
    ■■■ Path B: Wi-Fi direct → HTTPS → lot-systems.com
               Fallback only, used when no paired companion app is reachable.
```

Both paths terminate at the same backend contract (§02). The device firmware does not know or care which path is active — that decision is made by the companion app / device based on radio availability, per firmware §05.

01 — Pairing

The device has no keyboard and no way to enter a Wi-Fi password or log in. Pairing happens once, through the companion app, the same way a smartwatch pairs:

1. Companion app (already authenticated as a LOT® user via the existing session/cookie flow — **src/server/models/session.ts**) scans BLE, finds an unpaired LOT® Computer advertising its device ID.
2. App calls **POST /api/devices/pair** (**new endpoint, see §03**) with the device ID, under the user's existing authenticated session.
3. Server mints a **device-scoped token** (see §03) and returns it once.
4. App relays the token to the device over BLE. The device stores it in its OTA-protected settings partition. The token is never re-issuable through the device itself — only through an authenticated companion-app session, so a stolen device cannot be re-paired to a stranger's account without also compromising that stranger's LOT® login.

02 — The LOT® API Connector Contract

Two calls only. This is intentionally the entire surface area.

Receive: notification pull (or push, transport-dependent)

```
GET /api/devices/:deviceId/notifications/next
Authorization: Bearer <device-scoped token>

200 → { text: "Coffee time!", messageId: "...", issuedAt: "..." }
```

```
204 → no message pending
```

Poll interval is set by the companion app (typically 60-300s while phone is reachable over BLE) or, on the Wi-Fi-direct fallback path, a longer interval matched to the firmware power budget (§05, firmware doc). A future version may upgrade this to a push channel (webhook-to-BLE via the companion app, or MQTT over Wi-Fi) once volume justifies the added infrastructure — v1 ships with poll because it is simpler to audit and does not require a new always-on connection on the server side.

Respond: the Copy button

```
POST /api/logs
Authorization: Bearer <device-scoped token>
Body: {
  text: "Copy",
  event: "device.copy",
  metadata: { deviceId, messageId, tapTimestamp }
}
```

```
200 → { id, userId, text, event, metadata, context, createdAt }
```

This is **the same /api/logs endpoint the web app already uses** (**src/server/routes/api.ts**, **fastify.post('/logs', ...)**), not a new parallel table. The Copy tap lands in the person's existing Log tab, indistinguishable in storage from a log entry typed on the website — only **event: "device.copy"** and the **metadata.deviceId** distinguish its origin for display/filtering purposes.

Session-compression weather/presence records from the firmware's compression buffer (firmware doc §03) ride the same **/api/logs** call with **event: "device.session"** and the compressed payload in **metadata**, rather than opening a third endpoint — keeping the connector to two calls total.

03 — Required Backend Additions (Roadmap Item, Not Yet Built)

The device cannot hold a browser session cookie. Today's auth (**JWT_COOKIE_KEY**, **src/server/models/session.ts**) is built for a browser. Two small additions are needed before §01–§02 can go live — flagged here explicitly as build work for the roadmap, not assumed to already exist:

1. **Device model** — mirrors the existing **Session** model's shape (**token** PK, **userId** FK, **createdAt**, **expiresAt**, **lastUsedAt**) plus **deviceId** (hardware-burned identifier) and **pairedAt**. Same prune-on-expiry pattern as **Session.pruneExpired()**.
2. **Bearer-token auth middleware** for the two routes in §02, parallel to the existing cookie-session middleware, scoped so a device token can only ever call **GET /notifications/next** and **POST /logs** for its own paired **userId** — never the full authenticated-user API surface a browser session gets.

This is a contained, reviewable change (one new model, one new middleware, two route guards) — sized to land in a single PR before the first pilot unit ships.

04 — Companion App / "LOT Bridge"

A minimal app (mobile or desktop menu-bar utility) whose entire job is:

- Hold the BLE pairing relationship (§01)
- Relay poll requests / Copy events between the device and the platform when the device has no direct Wi-Fi reach
- Surface OTA availability to the user for confirmation (firmware §06) — OTA is pulled by the device but the human sees "Firmware update available" in the Bridge app first, consistent with the platform's human-gate-on-consequential-action principle already documented for server infrastructure (**docs/technical/LOT-NODE-0-RIG-SPEC.md**, §04).

The Bridge app has no independent UI beyond pairing + OTA consent. It is not a second dashboard — the dashboard is lot-systems.com, same as it is today.

05 — What This Connector Explicitly Does Not Do

- Does not create a new inbox, new notification table, or new user-facing surface. Notifications are authored by the existing QI·46 / Memory Engine pipeline on the platform; this document only specifies how they reach a physical object.
- Does not give a device token access to anything beyond its own two routes for its own paired user.
- Does not store platform credentials on the device. The device only ever holds the narrow, revocable device-scoped token from §01/§03.
- Does not require the device to be online continuously — a missed poll or a failed Copy sync is retried once (firmware §03) and dropped, not queued indefinitely.

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